

Muhammad Akbar Suharbi

Data Analyst | Data Scientist | Data Engineer

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SUMMARY

Data & AI Engineer with a strong foundation in computer engineering and business management, combining 7+ years of hands-on experience in data analysis, system development, and IT infrastructure across banking and enterprise sectors. Proven ability to translate complex data into actionable insights and build scalable systems including a tax automation system serving 15,000+ employees that reduced processing time by 97%, and managing IDR 350B funding portfolio at PT Bank Rakyat Indonesia. Currently advancing expertise in Machine Learning and AI Engineering through Hacktiv8 intensive program, with practical experience in Neural Networks, computer vision, and IoT-based data processing. Passionate about engineering intelligent data pipelines and deploying ML models that drive real business value.

EDUCATION

Hacktiv8 <i>Data Science & AI Machine Learning</i>	Jakarta, Indonesia May 2026 – July 2026
Universitas Airlangga Surabaya <i>Master of Management Business (GPA 3.64/4.00)</i>	Surabaya, Indonesia 2020 – 2022
Universitas Dinamika Surabaya <i>Bachelor of Computer Engineering (GPA 3.81/4.00)</i>	Surabaya, Indonesia 2015 – 2019

WORK EXPERIENCE

PT Bank Rakyat Indonesia <i>IT Staff</i>	Surabaya, Indonesia 2021 – 2026
Trusted by branch manager to transform raw business data across branch, sub-branch, and unit offices into actionable insights for strategic decision-making. Responsible for monitoring IT infrastructure including network, servers, CCTV, ATM/CRM, EDC, and end-user devices while coordinating with head office and vendors. - Transformed raw business data into strategic insights for branch performance evaluation - Maintained IT infrastructure ensuring minimal downtime across supervised units - Achieved 100% ATM/CRM reliability through proactive monitoring and vendor coordination - Maintained 95% EDC reliability through merchant monitoring and troubleshooting - Ensured CCTV monitoring and backup availability across operational units	
PT Arina Multikarya <i>IT Support and Developer</i>	Surabaya, Indonesia 2020 – 2021
Responsible for developing internal business applications, maintaining server and network infrastructure, and implementing workforce monitoring systems used nationwide. Built tax processing system and GPS-based attendance system to improve operational efficiency. - Developed PPh21 tax system for 15,000 employees reducing processing time from 1 day to 30 minutes - Built GPS-based attendance system used by 15,000 employees across Indonesia - Implemented leave, sick, and permission tracking features for workforce monitoring - Maintained server reliability with daily monitoring and preventive maintenance - Developed IT asset inventory system minimizing asset loss	

PT Total Quality

Full Stack Developer

Surabaya, Indonesia

2019 – 2020

Worked closely with CEO to develop web and mobile applications for agenda management and Change Agent ranking across corporate clients while managing company profile websites. Developed real-time brainwave monitoring integration to support data-driven motivation treatment.

- Delivered platform used by 37 corporate clients
- Increased workforce productivity by 25% and reduced operational cost by 8%
- Developed real-time brainwave monitoring application
- Led development team of 3 engineers

SKILLS

- **Database & Data Warehousing:** MySQL, Microsoft SQL Server, PostgreSQL, BigQuery, MongoDB, Data Warehouse Design, Dimensional Modeling, Data Normalization, Redshift
- **ETL & Data Pipeline:** ETL Development (script-based), Data Integration, Data Migration, Data Quality & Anomaly Detection, Apache Airflow (workflow orchestration), API Integration (RESTful APIs, FastAPI), Hadoop, Redis, Great Expectations, Kafka, Debezium
- **Programming Languages:** Python, SQL, C#, C/C++, PHP, JavaScript, Golang, Next.js
- **Data Visualization / BI:** Tableau, Power BI, Kibana, Microsoft Excel, Microsoft PowerPoint
- **Tools / Platforms:** Git, Docker, Anaconda, Jupyter Notebook, Visual Studio Code, Kubernetes
- **Python Libraries:** Pandas, NumPy, OpenCV, Matplotlib, Seaborn, Scikit-Learn, PyTorch, TensorFlow, Apache Spark, FastAPI, Selenium, BeautifulSoup
- **AI & Machine Learning:** Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), Large Language Models (LLMs), Elasticsearch, RAG, DAG
- **Collaboration & Documentation:** Technical documentation, cross-functional communication with head office and vendor teams, stakeholder reporting
- **Data Modeling Concepts:** Star Schema, Snowflake Schema, Fact & Dimension Tables, Data Normalization

PROJECTS

Smart Pharma Inventory Intelligence

2026

Project Description: Built an end-to-end MLOps and data pipeline for pharmaceutical inventory management using a Medallion Architecture (Bronze/Silver/Gold) orchestrated with Apache Airflow. Developed demand forecasting (Prophet) and expiry risk prediction models to optimize inventory decisions. Designed two LLM-powered AI Agents using Google Gemini API: a procurement recommendation agent with rule-based validation to prevent hallucinations and a function-calling chatbot for secure, data-driven warehouse analytics. Deployed a multi-page Streamlit application with interactive dashboards and conversational analytics.

Technology / Tools: Python, Apache Airflow, Docker, PostgreSQL, MinIO, Great Expectations, Prophet, Scikit-learn, XGBoost, LightGBM, Google Gemini API, Streamlit, Plotly, SQLAlchemy, Pandas, NumPy, Git/GitHub, LLM, AI Agent.

RetailIQ: Automated Revenue & Profit Analytics Pipeline

2026

Project Description: Built an end-to-end data pipeline for retail revenue and profit analysis, orchestrated with Apache Airflow across three scheduled tasks covering extraction from PostgreSQL, cleaning, and loading into Elasticsearch. Validated data quality with Great Expectations before it reached the pipeline, then designed an interactive Kibana dashboard with six visualizations breaking down revenue and profit by product category, customer segment, region, and account manager, giving the Sales and Finance teams a single reference for pricing and resource allocation decisions. Containerized the full stack with Docker Compose for reproducible deployment.

Technology/Tools: Python, Apache Airflow, PostgreSQL, Elasticsearch, Kibana, Great Expectations, Docker, psycpg2, Pandas, Git/GitHub.

Automatic Indonesian News Classification using Deep Learning (ANN)

2026

Project Description: Designed and deployed an end-to-end NLP classification system for 32,318 Indonesian news headlines, automatically categorizing articles into five editorial topics using an ensemble of deep learning models. Built a reproducible text preprocessing pipeline featuring Indonesian text normalization, tokenization, class imbalance handling, and leakage-safe train-validation-test splitting. Improved model performance by combining three TextCNN architectures with a Word2Vec-pretrained TextCNN through soft voting, achieving 89.29% F1-Macro and 89.89% test accuracy. Productionized the solution as a multi-page Streamlit application with batch inference, confidence-based review, EDA dashboard, and model serialization for real-world editorial workflows.

Technology / Tools: Python, TensorFlow/Keras, Gensim (Word2Vec), Scikit-learn, Pandas, NumPy, Sastrawi, Matplotlib, Seaborn, Streamlit, Git/GitHub.

Investor Risk Profile Classification for MiFID II Compliance

2026

Project Description: Built an end-to-end 4-class classification system (Conservative, Income, Balanced, Aggressive) on 21,000+ investor records to support MiFID II-driven suitability assessments for Relationship Manager and Compliance teams. Engineered a leakage-safe preprocessing pipeline (stratified train-test split, IQR-based outlier capping, ColumnTransformer with StandardScaler and hybrid One-Hot/Ordinal encoding) after validating feature relevance through ANOVA F-tests and Chi-Square tests, and resolved multicollinearity via correlation-driven feature selection. Benchmarked 5 algorithms (KNN, SVM, Decision Tree, Random Forest, XGBoost) via 5-fold cross-validation, then optimized the best performer through RandomizedSearchCV, improving F1-Macro by ~1.3 points over baseline while ensuring zero high-risk misclassifications (e.g., no Conservative investor ever misclassified as Aggressive), a critical safety property for regulatory risk tooling. Deployed the final model as an interactive Streamlit web app on Hugging Face Spaces with a custom preprocessing transformer module for production-safe model serialization.

Technology / Tools: Python, Pandas, NumPy, Scikit-learn, XGBoost, SciPy (Stats), Matplotlib, Seaborn, Streamlit, Hugging Face Spaces, Git/GitHub.

Pizza Sales Performance 2025 & Q1 Projection 2026

2026

Project Description: Analyzed a large-scale dataset of over 50,000 historical retail transactions to identify operational inefficiencies and revenue-generating patterns to support a 20% Q1 growth target. Engineered a robust data cleansing pipeline in Python to resolve critical time-delta anomalies and formatted structured CSV inputs for Tableau. Conducted descriptive statistics (Skewness & Kurtosis) and inferential hypothesis testing (Independent T-Test) to scientifically validate a high-margin "Golden Hour" during lunch peaks, creating an interactive Executive Dashboard that transformed raw data into actionable inventory and promo strategies.

Technology / Tools: Python, Pandas, NumPy, SciPy (Stats), Matplotlib, Seaborn, Tableau Desktop, Microsoft Excel.

Medical Report CRR

2021

Project Description: Developed a web-based medical report platform designed to streamline and automate the processing of credit recovery data into an interactive summary dashboard. Integrated MySQL databases with a CodeIgniter and Bootstrap frontend to handle data migration, reducing manual reporting time by **70%** and improving data visualization accuracy for stakeholders.

Technology / Tools: PHP, CodeIgniter, MySQL, JavaScript, Bootstrap, CSS, Microsoft Excel.

CERTIFICATIONS

HackerRank

Python (Basic)

Certificate: ID-84c4960-ff2b-43f5-b6e9-a7e8e07f3a74

2026

Badan Nasional Sertifikasi Profesi (BNSP)

Wakil Perantara Perdagangan Efek (WPPE)

Certificate: 66142241270004492023

2023

KOMINFO

Cyber Security Analyst

Certificate: 05106073151-31/TA.DTS/BLSDM.KOMINFO/2021

2021

SAP – Business Processes in Financial Accounting – ACM010

Certificate: Credential ID 550106107

2020